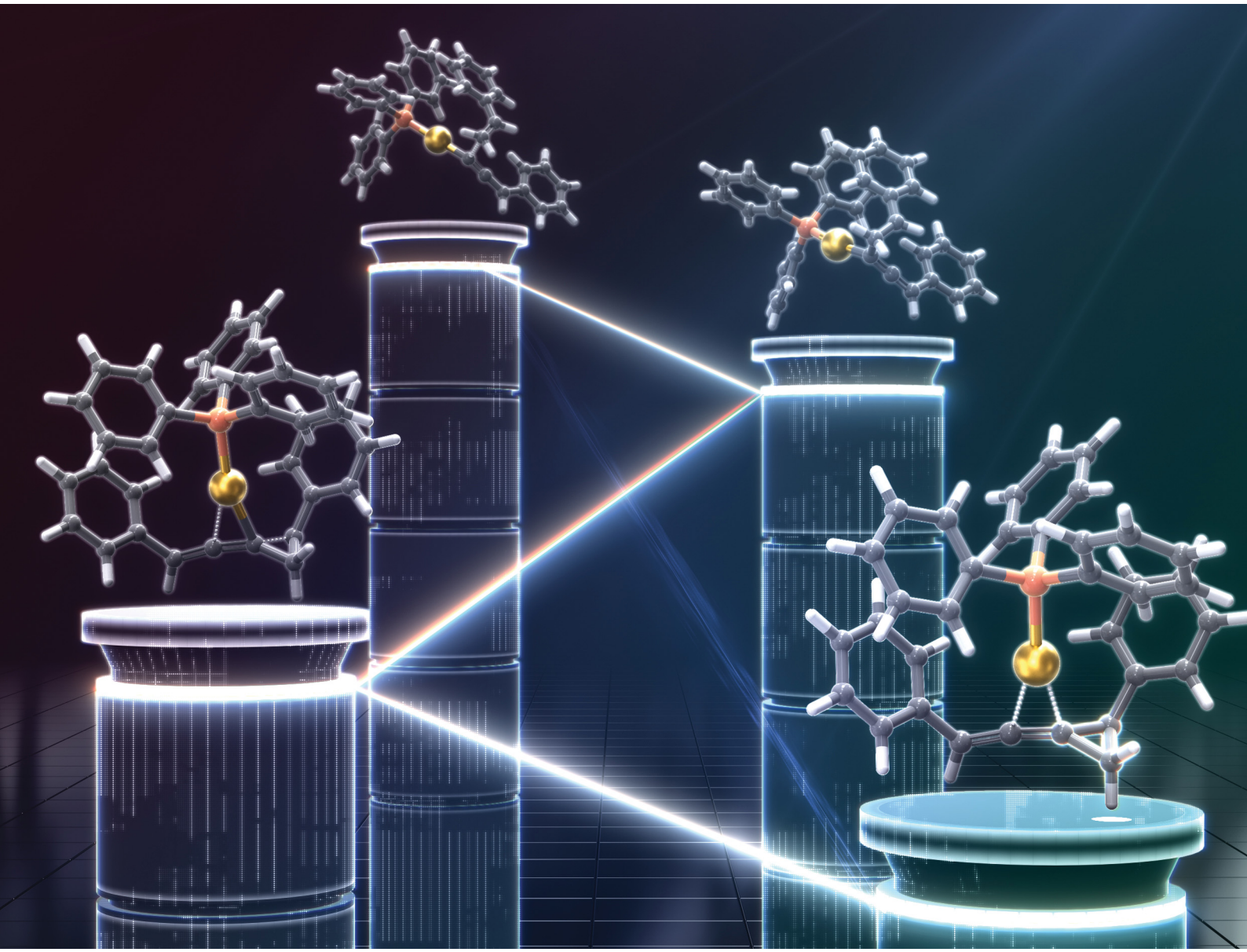
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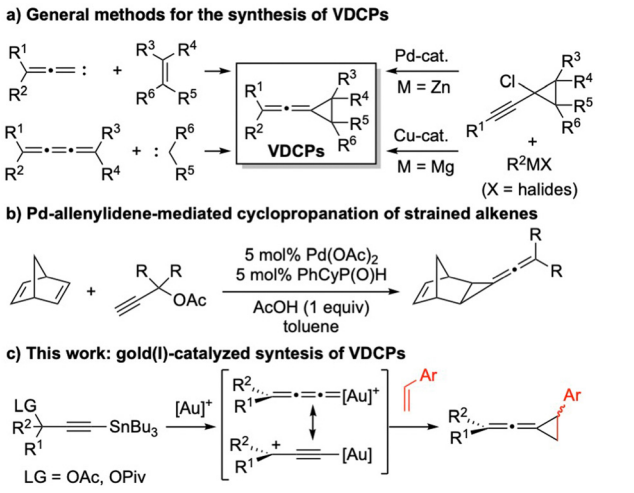
Synthesis of vinylidenecyclopropanes *via* gold(I)-catalyzed cyclopropanation of vinyl arenes with γ-stannylated propargyl esters†

Hiroto Mori,[a] Yusuke Ono,[a] Shota Nakagawa,[a] Sota Akima,[a] Miki Murakami,[b] Toshinobu Korenaga, *[cd] Tadashi Nakaji-Hirabayashi, [be] Mayumi Kyogoku[b] and Yoshikazu Horino *[a]

The reaction of γ-stannylated propargyl esters in the presence of a cationic gold(I) catalyst affords vinylidenecyclopropane derivatives as a mixture of diastereomers. The *cis*-geometry of the alkenes is almost entirely retained in the product. DFT calculations suggest the involvement of the gold(I)-stabilized propargyl cation as a resonance form of gold(I)-coordinated allenylidene species.

Cyclopropanes are important functional molecules found in numerous natural products,[1] bioactive molecules,[2] and ligands in catalysis,[3] and are a platform for further transformations.[4] In particular, vinylidenecyclopropanes (VDCPs) are synthetically valuable molecules[5] owing to their remarkable properties compared to other simple cyclopropanes and alkylidenecyclopropanes, including an inherently higher strain energy than simple cyclopropanes[6] as well as a synthetically versatile allene moiety.[7]

VDCPs are typically synthesized by the [2+1] cycloaddition of carbenes with unsaturated compounds (Scheme 1a).[8] However, the synthetic utility of this method is limited by its requirement for a strong base and the additional steps to prepare 1,2,3-butatrienes.[9] Another approach to the synthesis of VDCPs involves the Pd(0)[10]- or Cu(I)[11]-mediated substitutions of 1-alkynylcyclopropyl chlorides; however, the synthesis of 1-alkynylcyclopropyl chlorides requires multiple steps. In contrast, transition metal allenylidene species are sufficiently stable for use in organic synthesis and are readily generated by alkyne activation.[12] The cyclopropanation of alkenes through transition-metal allenylidene species is therefore a straightforward approach to the synthesis of VDCPs. Transition metal allenylidene complexes typically react *via* a similar mechanism to that of transition metal vinylidene complexes; thus, nucleophilic additions can occur at both the α- and γ-positions of the allenylidene complex depending on the nature of the metal and the nucleophile. Although several allenylidene-mediated C–C bond forming reactions at the γ-position have been devised using various transition metals, including Cu and Ru, studies investigating the reactivity of the α-position of transition metal allenylidene species have lagged significantly behind.[13] Buono's seminal work on the [2+1] cycloaddition of strained alkenes with palladium allenylidene species is the only example of a catalytic C–C bond-forming reaction at the α-position (Scheme 1b).[14] Gold catalysis has recently emerged as a powerful tool for three-membered ring-forming reactions;[15] thus, we hypothesized that as yet unexplored gold(I)-coordinated allenylidene species will provide a novel route



Scheme 1 Synthetic routes to VDCPs.

[a] *Department of Applied Chemistry and Bioscience, Chitose Institute of Science and Technology, Chitose, 066-8655, Japan. E-mail: y-horino@photon.chitose.ac.jp*
[b] *Graduate School of Science and Engineering, University of Toyama, Toyama, 930-8555, Japan*
[c] *Department of Applied Chemistry and Bioscience, Iwate University, Morioka, 020-8551, Japan. E-mail: korenaga@iwate-u.ac.jp*
[d] *Soft-Path Science and Engineering Research Center (SPERC), Iwate University, Morioka, 020-8551, Japan*
[e] *Faculty of Engineering, Academic Assembly, University of Toyama, Toyama, 930-8555, Japan*

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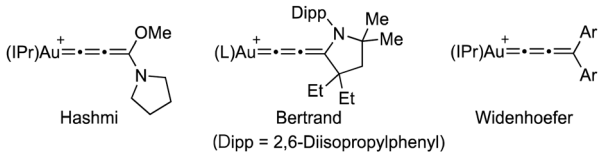


Fig. 1 Representative gold(I) allenylidene complexes. Counter anions are omitted for clarity.

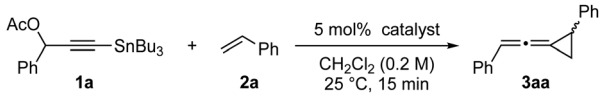
to the catalytic synthesis of VDCPs. Despite considerable progress in the development of gold-catalyzed alkene cyclopropanation reactions,[16] alkene cyclopropanation for the synthesis of VDCPs *via* a gold(I)-coordinated allenylidene species is underdeveloped owing to a lack of synthetic methods that afford such intermediates. Since the pioneering study on the preparation of a gold(I) (alkoxy)(amino)allenylidene complex by the Hashmi group,[17] Bertrand synthesized gold(I) mono- and bis-allenylidene complexes derived from a cyclic (alkyl)(amino)carbene (Fig. 1).[18] Later, Widenhoefer first synthesized and structurally characterized gold(I) diarylallenylidene complexes lacking heteroatom stabilization.[19] Despite these seminal studies, only stoichiometric carbon–heteroatom-bond-forming reactions have been achieved.[19] The exploration of catalytic C–C bond forming reactions *via* a gold(I)-coordinated allenylidene species or its resonance form, the gold(I)-stabilized propargyl cation, has therefore emerged at the forefront of current research.[20] We herein report the synthesis of VDCPs *via* the gold(I)-catalyzed cyclopropanation of vinyl arenes (Scheme 1c).

The reaction conditions were optimized based on the reaction of **1a** and styrene (**2a**). In the presence of $(Ph_3P)AuCl/AgSbF_6$ (5 mol%) in CH_2Cl_2 (0.2 M) at 25 °C, the reaction afforded the desired VDCP **3aa** in 57% yield as a 2 : 1 mixture of diastereomers (Table 1, entry 1). Reducing the amount of **2a** diminished the yield of **3aa** (entry 2). Screening of the counter ion, ligand, and gold complex[21] showed that the combination of $(Ph_3P)AuCl$ and $AgSbF_6$ exhibited high catalytic activity to give **3aa** (entries 1 and 3–6). The choice of ligand did not significantly influence the diastereoselectivity of the reaction. However, prolonging the reaction time led to decreased diastereoselectivity and partial decomposition of **3aa**, indicating that the gold(I)-mediated isomerization of **3aa** is involved (entry 7).[22] In addition, CH_2Cl_2 was the optimal solvent.[23] Finally, **3aa** was obtained in 68% yield by reducing the concentration (entry 8). No reaction occurred in the absence of $(Ph_3P)AuCl$ (entry 9).

The scope of the reaction was examined under the optimal conditions using various vinyl arenes (Scheme 2). The gold(I)-catalyzed cyclopropanation reaction of halogenated styrenes **2b** and **2c** with a chloro or bromo group at the *para* position gave **3ab** and **3ac** in 52% and 40% yields, respectively. The cyclopropanation reaction is sensitive to the electronic effects of the substituents on the phenyl ring of styrene. For example, although *p*-MeO- and *p*-CF_3-substituted styrenes were not compatible, the cyclopropanation of *o*-vinylnaphthalene and *o*-triisopropylsilyl (TIPS)-ethynylated styrene proceeded to afford **3af** and **3ag** in 43% and 56% yields, respectively. In addition, the chemoselective reaction of the styrene double bond of *o*-(3-butenyl)styrene yielded **3ah** in 57% yield. Furthermore, the present method was also applicable for α-substituted styrenes, such as α-methyl, α-phenyl, and α-cyclopropylstyrenes, affording **3ai–ak** in moderate to good yields. Interestingly, cyclopropanation of *cis*-stilbene did not proceed under the optimized conditions; however, *trans*-stilbene afforded the desired product **3al** in 44% yield. Similarly, cyclopropanes **3am** and **3an** were obtained from the cyclic alkenes such as indene and 1,2-dihydronaphthalene. The applicability of the reaction conditions at larger scales was demonstrated by the synthesis of **3am** in 80% yield. In all cases, VDCPs were obtained as a mixture of diastereomers. In contrast, although aliphatic alkenes and benzofuran are known to participate in the gold(I) carbene-mediated alkene cyclopropanation,[16,23] performing the reaction using them failed in the present system.

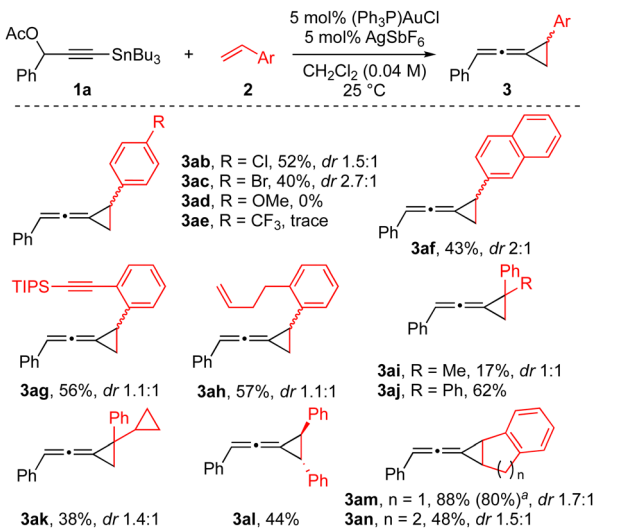
We also examined the substrate scope to assess the influence of the electronic nature of the propargyl substituent on the

Table 1 Optimization of the reaction[a]



Entry	Catalyst	3aa[b] (%)	d.r.
1	$(Ph_3P)AuCl/AgSbF_6$	57	2 : 1
2[c]	$(Ph_3P)AuCl/AgSbF_6$	27	2 : 1
3[d]	JohnPhosAuSbF$_6$·MeCN	57	1.3 : 1
4	$(4\text{-}CF_3C_6H_4)_3PAuCl/AgSbF_6$	50	1.8 : 1
5	$(4\text{-}MeOC_6H_4)_3PAuCl/AgSbF_6$	47	1.8 : 1
6[e]	(IPr)AuCl/AgSbF$_6$	14	1.2 : 1
7[f]	$(Ph_3P)AuCl/AgSbF_6$	39	1.2 : 1
8[g]	$(Ph_3P)AuCl/AgSbF_6$	68	2 : 1
9	$AgSbF_6$	0	

[a] Reactions were performed using **1a** (0.2 mmol), **2a** (2 mmol), Au (5 mol%), Ag (5 mol%), and solvent (1 mL) at 25 °C. [b] Isolated yield. [c] **2a** (1 mmol) was used. [d] Reaction time: 7 h. [e] Reaction time: 24 h. [f] Reaction time: 12 h. [g] 0.04 M concentration. JohnPhos = (2-biphenylyl)di-*tert*-butylphosphine. IPr = 1,3-bis(2,6-diisopropylphenyl)imidazol-2-ylidene.

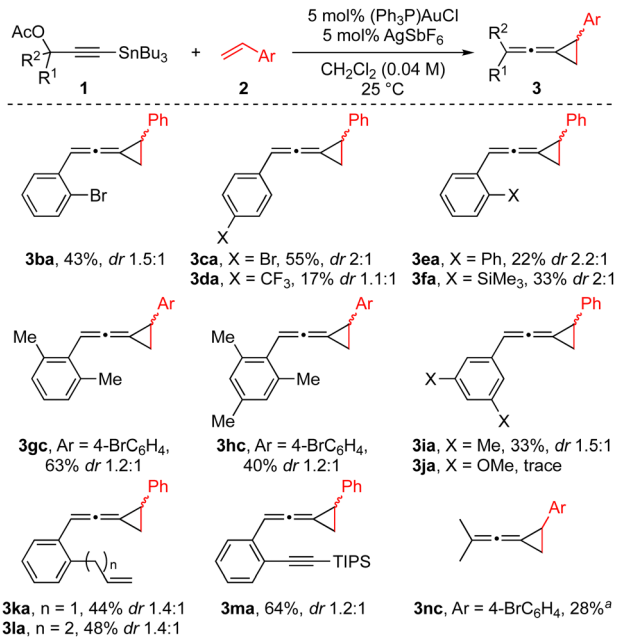


Scheme 2 Substrate scope of gold(I)-catalyzed cyclopropanation of vinyl arenes. [a] **1a** (1 mmol) was used.

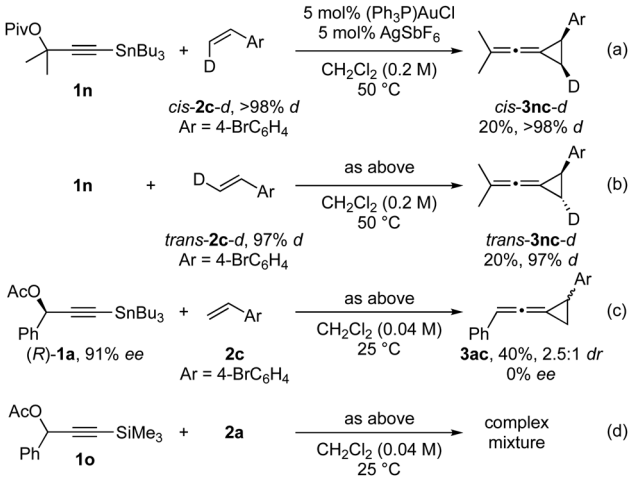
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reaction (Scheme 3). Phenyl groups at the propargylic position bearing bromo (**2b** and **2c**), trifluoromethyl (**2d**), phenyl (**2e**), trimethylsilyl (**2f**), and methyl (**2g–i**) groups at the *ortho*-, *meta*-, and *para*-positions yielded the corresponding VDCPs **3ba–ia** in low to good yields. The major isomer of **3hc** was determined to be *syn*.[23] The cyclopropanation reaction was found to be sensitive to the presence of oxygen-containing functional groups. For instance, the 3,5-dimethoxyphenyl-substituted substrate **1j** exhibited significantly lower reactivity than substrates without oxygen-containing groups, yielding only trace amount of the desired product **3ja**. The reaction conditions also tolerated unsaturated substituents, including *o*-allyl, *o*-homoallyl, and *o*-(TIPS)-ethynyl groups, to afford **3ka–ma**, and alkene and alkyne remained intact. Although the desired product **3nc** could not be obtained using 1,1-dimethyl-3-(tributylstannyl)-propargyl acetate, changing from an acetoxy group to a pivaloyl group furnished **3nc** in 28% yield.

To investigate the stereochemical course, we next achieved the reaction of **1n** with the *cis*- and *trans*-β-monodeuteriostyrene derivatives *cis*-**2c**-*d* and *trans*-**2c**-*d* (Scheme 4a and b).[23] The reaction was stereospecific with respect to the alkene configuration. For example, the reaction of **1n** with *cis*-**2c**-*d* proceeded with complete retention of the double bond configuration. Similarly, the alkene geometry of *trans*-**2c**-*d* was also retained completely in the product. In addition, *cis*- and *trans*-**2c**-*d* were recovered without any isomerization after the reaction with **1n**. Moreover, a control experiment was conducted to clarify the mechanism. Gold(I)-catalyzed reaction of **1a** (91% ee) with **2c** furnished **3ac** in 40% yield with complete loss of enantiomeric excess (Scheme 4c). Furthermore, reaction of **1o**, which possesses a $SiMe_3$ group instead of $SnBu_3$, yielded a complex mixture (Scheme 4d).

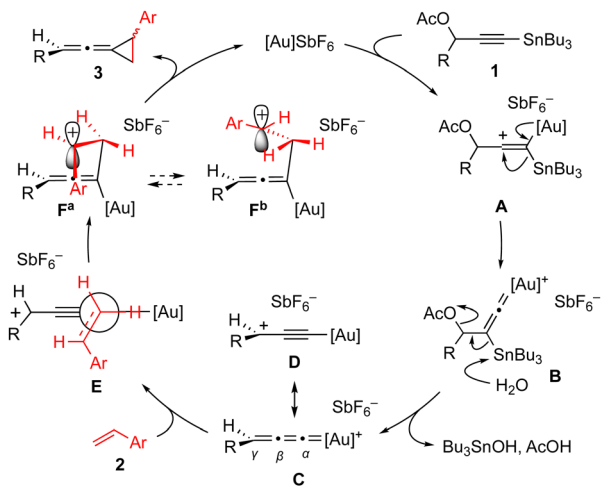


Scheme 3 Scope of the cyclopropanation of different vinyl arenes **2** with **1**. [a] Reaction was performed at 50 °C in CH_2Cl_2 (0.2 M) using a pivaloyl group instead of an acetoxy group.



Scheme 4 Mechanistic studies of gold(I)-catalyzed cyclopropanation of vinyl arenes.

A tentative reaction mechanism based on the above results and our previous study[24] is presented in Scheme 5. The complexation of **1** with the gold(I) complex generates the vinyl cation intermediate **A**, which is efficiently stabilized by complexation, hyperconjugation with the β-stannyl substituent, and conventional neighboring group participation of the acetoxy group. π-backdonation from gold(I) induces rearrangement of the stannyl group to afford the gold(I) carbene intermediate **B**, which would then produce the gold(I)-coordinated allenylidene species **C** *via* a bimolecular *anti*-specific elimination.[25] DFT calculations were performed to obtain further details on the cyclopropanation mechanism.[23] Although the cyclopropanation of styrene with gold(I)-coordinated benzylidene species has been proposed to occur *via* a concerted mechanism,[26] our calculations show that the cyclopropane ring is constructed by a stepwise mechanism. The vinyl arene **2** attacks the Cα carbon of the gold(I)-stabilized propargyl cation species **D** as shown in **E** to give the carbenium intermediate $\mathbf{F}^{a}$ (1st step). Electrophilic cleavage of the $[Au]^+$ *via* the interaction of the back lobe of the Au–Cα bond with the



Scheme 5 Plausible reaction mechanism. [Au] = $(Ph_3P)Au$.

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benzylic cation affords **3** (2nd step).[26] The activation free energies of both steps indicated that the 1st step is slower than the 2nd step. In addition, although there is a possibility of equilibrium between $\mathbf{F}^{a}$ and $\mathbf{F}^{b}$, higher activation free energies than the energies of the 2nd step make this equilibrium impossible.[23] Therefore, the reaction proceeds straightforwardly without branching along the way, which is consistent with the experimentally observed stereochemical outcome, as shown in Scheme 4a and b.

In conclusion, we have developed a mild and general method for the synthesis of VDCPs *via* the gold(I)-catalyzed cyclopropanation of vinyl arenes. This method represents the first catalytic method for the synthesis of VDCPs *via* the cyclopropanation of vinyl arenes. Further exploration of new methodology *via* a gold(I)-stabilized propargyl cation species is currently underway in our laboratory.

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Data availability

The ESI† includes the experimental procedures, compound characterization data, DFT calculations, and NMR spectra.

Conflicts of interest

There are no conflicts of interest to declare.

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